NED University of Engineering and Technology, Karachi

Computer Science and Information Technology

COMPLEX COMPUTING PROBLEM (CCP)
PROJECT SUBMISSION



Knowledge-Based Systems (CT-359)

Project Title	Intelligent Knowledge-Based System for Industrial Equipment Fault Diagnosis and Maintenance Decision Support
Submission Date	01/11/2026

Student Details

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Project Metadata:

- **Project Title:** Intelligent Knowledge-Based System for Industrial Equipment Fault Diagnosis and Maintenance Decision Support
- **Submission Date:** October 27, 2026

1. Project Instructions

- Complete all sections using the prescribed template, replacing bracketed placeholders with project details.
- Submit source code, dataset/knowledge base files, report, and presentation before the deadline.
- Implementation must be demonstrable and executable during the viva.
- All assumptions, external sources, tools, and libraries must be explicitly stated and cited.

2. Problem Statement

- **CLO 1 (C2, PLO-2):** Represent industrial equipment knowledge (equipment, components, symptoms, faults, causes, maintenance actions) using an appropriate KR formalism.
- **CLO 2 (C4, PLO-3):** Analyze industrial fault diagnosis by identifying incomplete requirements, stakeholder needs, relationships, and operational constraints.
- **CLO 3 (C5, PLO-4):** Evaluate KR methods and inference algorithms, justifying choices based on accuracy, explainability, safety, cost, and system requirements.

3. Problem Description / Project Context

A manufacturing facility operating **Electric Motors**, **Water Pumps**, and **Air Compressors** faces diagnostic delays due to fragmented expertise scattered across manufacturer guides (Atlas Copco, Kaeser, Sulzer), service manuals (Haas CNC, Punjab Water Supply), and engineering literature (EASA Motor Overheating, Vibration, Noise). Operators observe surface symptoms (overheating, vibration, noise, pressure drops, leakage) without immediate root-cause clarity.

The organization requires a **Knowledge-Based System (KBS)** to unify this heterogeneous knowledge into a structured formalism, infer candidate faults under incomplete or multi-symptom observations, reconcile conflicting stakeholder priorities (safety vs. downtime vs. maintenance cost), recommend actionable maintenance procedures, and provide a transparent reasoning trace.

4. Problem Analysis

4.1 Problem Understanding

Industrial fault diagnosis is an ill-structured domain conforming to Seoul Accord / NCEAC Complex Computing Problem Attributes (**CPA-2, CPA-3, CPA-9**):

1. **Many-to-Many Mappings:** The symptom `Unusual noise` indicates bearing wear in motors (R3), hydraulic cavitation in pumps (R5), or loose hardware in compressors (R9).
2. **Uncertainty & Incomplete Evidence:** Shop-floor operators observe qualitative symptoms before invasive tear-downs or sensor instrumentation.
3. **Coupled Decision-Making:** Candidate faults must be tied directly to safe maintenance actions and prioritized to prevent catastrophic failure.

4.2 Stakeholders and Their Requirements

Stakeholder	Role	Core Requirement
Field Technicians	Floor operators reporting machine abnormalities.	Intuitive multi-symptom input, actionable inspection steps, and transparent reasoning.
HSE / Safety Engineers	Workplace hazard prevention and regulatory compliance.	Priority escalation for thermal hazards (R1, R7) and high-pressure leaks (R6).
Production Managers	Line continuity and output quotas.	Rapid fault isolation to minimize unplanned downtime.
Cost Controllers	Maintenance budgeting and spare-part logistics.	Non-invasive preliminary checks before ordering costly component overhauls.
Knowledge Engineers	System maintenance and scalability.	Clean separation of knowledge base (`.csv`) from the Python inference engine.

4.3 Constraints and Conflicting Requirements

- **Safety vs. Production Continuity:** Production demands continuous operation, while safety mandates immediate stoppage for thermal runaway (R1, R7) or pressurized pipe leaks (R6). The system resolves this via explicit `High` priority tagging.
- **Diagnostic Cost vs. Completeness:** Disassembling equipment is costly; the KBS recommends non-destructive first-line checks (e.g., checking filters and suction lines) before invasive overhauls.
- **Incomplete Observations:** To avoid blocking operators when precision telemetry is unavailable, the KBS accepts high-level qualitative observations and presents candidate hypotheses with an epistemic disclaimer.

4.4 Root Cause / Requirement Identification (CPA-9)

Requirements identified from the domain include multi-symptom input support, source traceability (`source_id`), and explicit advisory disclaimers. Root causes are grouped into:

1. **Thermal:** Blocked motor cooling vents (R1), compressor restricted airflow (R7).
2. **Mechanical/Dynamic:** Rotor imbalance/misalignment (R2), worn bearings (R3), loose hardware (R9).
3. **Fluid/Pneumatic:** Suction blockage (R4), pump cavitation (R5), pipe joint failure (R6), air leaks/clogged filters (R8).

4.5 Assumptions

- **Closed-World Scope:** The system covers three equipment types: Electric Motor, Water Pump, and Air Compressor, with 15 diagnostic and inference rules (R1–R15) stored in the CSV knowledge base.
- **Qualitative Operator Inputs:** Symptoms and additional conditions are entered by the operator through the interactive command-line interface.
- **Multi-Fault Coexistence:** All rules matching the selected equipment and reported symptoms can be activated, allowing multiple possible faults to be considered.
- **Multi-Level Reasoning:** The system uses additional evidence and inference rules to derive further possible fault conditions.
- **Cost Estimation:** Maintenance costs are hypothetical academic estimates in Pakistani Rupees (PKR), not verified market quotations.
- **Stakeholder Prioritization:** Stakeholder weights and impact scores are illustrative values used to demonstrate weighted decision ranking.
- **Certainty Factors:** Certainty factors are manually assigned academic support values and are not statistically validated probabilities.

- **Portability:** Diagnostic rules and source information are stored in local CSV files without requiring external server dependencies.
- **Source Verification:** Diagnostic information is linked to reference sources through source IDs. Verification status is maintained separately for each source.
    1.

5. Knowledge Representation and KBS Design

5.1 Selected Knowledge Representation Formalism

A Rule-Based Knowledge Representation approach was selected for the project. Diagnostic knowledge is stored in a structured CSV knowledge base, where each rule connects an equipment type and an observed symptom with a possible fault, possible cause, maintenance action, priority, and source ID.

The system uses IF-THEN-style diagnostic rules to identify possible faults based on the equipment and symptoms selected by the user. This approach was preferred because it is simple, understandable, explainable, and allows diagnostic knowledge to be updated separately from the Python inference program.

5.2 Knowledge Base / Rule Base (knowledge_base.csv)

The knowledge base contains 15 rules divided into two categories:

1. **Original diagnostic rules (R1–R9): Match equipment and observed symptoms to possible faults, causes, and maintenance recommendations.**
2. **Additional inference rules (R10–R15): Use confirmed additional conditions and parent fault hypotheses to derive further possible fault conditions.**

Rule ID	Equipment	Symptom (Evidence)
R1	Electric Motor	Overheating
R2	Electric Motor	Unusual vibration

Rule ID	Equipment	Symptom (Evidence)
R3	Electric Motor	Unusual noise
R4	Water Pump	Reduced water flow
R5	Water Pump	Unusual noise
R6	Water Pump	Water leakage
R7	Air Compressor	High temperature
R8	Air Compressor	Low pressure
R9	Air Compressor	Unusual noise

Additional inference rules:

Rule ID	Equipment	Additional Condition	Derived Conclusion	Parent Fault
R10	Electric Motor	Restricted ventilation observed	Possible restricted ventilation	Possible cooling problem
R11	Electric Motor	Inadequate cooling airflow observed	Possible inadequate cooling airflow	Possible cooling problem
R12	Water Pump	Restricted suction flow observed	Possible suction restriction	Possible water flow or pressure problem
R13	Water Pump	Low inlet water level observed	Possible insufficient water supply	Possible water flow or pressure problem

R14	Air Compressor	Restricted cooling airflow observed	Possible cooling airflow restriction	Possible cooling problem
R15	Air Compressor	Air leakage observed	Possible compressed air leakage	Possible air leak or pressure problem

5.3 Inference Mechanism

The proposed Knowledge-Based System uses forward chaining and a limited backward-chaining hypothesis verification mechanism.

A. Forward Chaining (Data-Driven Reasoning)

Forward chaining starts with the facts supplied by the user and applies matching rules to derive possible conclusions.

The process consists of:

1. **Equipment Selection:** The user selects Electric Motor, Water Pump, or Air Compressor.
2. **Symptom Extraction:** The system retrieves available symptoms for the selected equipment.
3. **User Input:** The user enters one or more observed symptoms.
4. **Initial Rule Matching:** The inference engine checks the knowledge base for diagnostic rules matching the selected equipment and symptoms.
5. **Initial Fault Identification:** Matching rules identify possible faults, causes, maintenance actions, and priorities.
6. **Additional Evidence Collection:** When a parent fault hypothesis is identified, the system asks the user to confirm relevant additional conditions.
7. **Additional Inference:** Rules R10–R15 activate when their required equipment, parent fault, and additional condition are satisfied.
8. **Fact Derivation:** The system adds the resulting conclusions to its working memory and calculates combined certainty-factor support where applicable.
9. **Result Generation:** The system displays activated rules, possible faults, confirmed conditions, derived facts, priorities, maintenance suggestions, and reasoning information.

The system therefore demonstrates multi-level reasoning by moving from initial symptoms to possible faults and then to more specific conclusions based on additional evidence.

B. Backward Chaining (Goal-Driven Hypothesis Verification)

Backward chaining begins with a possible fault hypothesis selected by the user.

The system:

1. Displays possible fault hypotheses generated during forward chaining.
2. Allows the user to select a hypothesis for verification.
3. Searches the knowledge base for a matching diagnostic or inference rule.
4. Checks whether the required original symptom or additional confirmed condition is available.
5. Displays whether the selected hypothesis is supported by the available evidence.

The implemented backward-chaining function performs limited hypothesis checking rather than general-purpose recursive theorem proving.

Both reasoning mechanisms improve transparency by showing the rules and evidence associated with the system's conclusions. Results represent possible faults, not confirmed equipment diagnoses.

5.4 Conflict Resolution and Decision Support Strategy

The system uses multiple strategies to support maintenance decision-making.

A. Rule Matching

Only rules associated with the equipment selected by the user are considered. Multiple matching rules can be activated when several symptoms are reported.

B. Safety Priority

Each rule has a predefined priority level, such as High or Medium. High-priority alerts are displayed as advisory warnings. The system does not automatically stop or control equipment.

C. Stakeholder-Based Weighted Priority Ranking

The system uses a weighted scoring method to demonstrate how different stakeholder concerns can contribute to maintenance prioritization.

The illustrative stakeholder weights are:

Stakeholder	Weight
Safety Officer	0.40
Production Manager	0.30
Maintenance Manager	0.20
Finance Manager	0.10
Total	**1.00**

Each stakeholder assigns an illustrative impact score from 1 to 5 for an activated rule.

The weighted priority score is calculated as:

Weighted Score = Σ (Stakeholder Weight × Impact Score)

Rules are then ranked according to their calculated weighted scores.

These values demonstrate a weighted ranking approach. They do not represent an automated negotiation or complete resolution of stakeholder conflicts.

5.5 Explanation Facility and System Architecture

The explanation facility provides transparency by displaying:

- Equipment and symptoms entered by the user.
- Activated diagnostic and inference rule IDs.
- Possible faults, causes, and maintenance recommendations.

- Additional conditions confirmed by the user.
- Derived facts generated through inference.
- Priority levels and safety advisory messages.
- Source IDs associated with activated rules.
- Backward-chaining hypothesis verification results.
- Certainty-factor support values.
- Estimated maintenance cost alternatives.
- Stakeholder-based weighted priority rankings.

Updated System Architecture

User (CLI) → Equipment and Symptom Input → Working Memory → Forward-Chaining Inference → Additional Evidence Collection → Multi-Level Rule Inference → Certainty-Factor Calculation → Maintenance Decision Support → Diagnostic Results and Explanation.

Additional functions provide backward-chaining hypothesis verification, maintenance cost estimation, safety alerts, and stakeholder-based weighted priority ranking.

6. Dataset / Knowledge Sources

Source ID	Source / Organization	Type
S1	Atlas Copco	Manufacturer Guide
S2	Haas Automation	Service Manual
S3	LG&CD Punjab	Municipal O&M Manual
S4	EASA	Technical Article
S5	EASA	Technical Article
S6	EASA	Technical Article
S7	Sulzer Ltd.	Manufacturer Guide

Source ID	Source / Organization	Type
S8	Kaeser Compressors	Manufacturer Guide

7. Implementation

Environment and Tools: Python 3.x, built-in `csv` module, `csv.DictReader`, and UTF-8-SIG encoding.

Implementation Components:

1. Equipment Selection Module
2. CSV Knowledge Base Loading and Validation
3. Dynamic Symptom Extraction
4. Forward-Chaining Inference Engine
5. Additional Evidence Collection
6. Multi-Level Inference Rules
7. Backward-Chaining Hypothesis Verification
8. Certainty-Factor Calculation
9. Maintenance Cost Estimation
10. Stakeholder-Based Priority Ranking
11. Safety Alert and Explanation Module

The knowledge base is maintained separately in CSV format, allowing diagnostic and inference rules to be updated without changing the main Python program.

The implementation uses hypothetical cost estimates, stakeholder impact scores, and certainty-factor values for academic demonstration.

8. Testing and Results

8.1 Test Cases and Sample Results

Test Case	Equipment Selected	Initial Symptom	Additional Evidence Confirmed	Expected Activated Rules	Status
TC-01	Electric Motor	Overheating	Restricted ventilation observed	R1, R10	PASS

TC-02	Water Pump	Reduced water flow	Restricted suction flow observed	R4, R12	PASS
TC-03	Air Compressor	High temperature	Restricted cooling airflow observed	R7, R14	PASS

The test cases demonstrate that the system can identify initial possible faults and activate additional inference rules when the required evidence is confirmed.

8.2 Additional Functional Testing

The following additional functions were tested during system execution:

- Equipment-specific symptom selection.
- Multiple symptom matching.
- Forward-chaining inference.
- Backward-chaining hypothesis verification.
- Maintenance cost estimation.
- Stakeholder-based weighted priority ranking.
- Certainty-factor display.
- High-priority safety advisory messages.
- Invalid input handling.

8.3 Results and Discussion

The system successfully demonstrates a rule-based approach to industrial fault diagnosis and maintenance decision support.

The forward-chaining mechanism identifies possible faults from operator-entered symptoms. Additional inference rules derive more specific conclusions when the required evidence is confirmed.

The backward-chaining function provides limited verification of selected fault hypotheses against available evidence.

Maintenance cost estimation presents hypothetical inspection, repair, and replacement alternatives in PKR. Stakeholder-based ranking calculates weighted scores to compare maintenance priorities. Certainty factors provide numerical support values for the system's conclusions.

The system provides an explanation of activated rules and associated evidence, improving the transparency of its recommendations.

The results are advisory and should not be treated as confirmed diagnoses or automatic equipment-control decisions.

9. Evaluation, Limitations and Conclusion

9.1 Evaluation of the Proposed Approach (CLO 3)

Knowledge Representation: The system represents equipment, symptoms, possible faults, causes, maintenance actions, priorities, and inference relationships through a structured CSV rule base.

Reasoning: Forward chaining supports symptom-driven inference, while the backward-chaining function provides limited hypothesis verification.

Explainability: Activated rules, evidence, derived facts, and source IDs provide a transparent reasoning trace.

Safety: High-priority rules generate advisory warnings. The system does not automatically stop equipment or control machinery.

Cost Evaluation: Hypothetical inspection, repair, and replacement estimates provide a basic comparison of maintenance alternatives.

Stakeholder Decision Support: Weighted stakeholder impact scores demonstrate a numerical ranking approach for maintenance priorities.

Uncertainty Representation: Manually assigned certainty factors provide numerical support values. These are illustrative and are not statistically validated probabilities.

Computational Efficiency: The lightweight Python implementation uses local CSV files and does not require external server infrastructure.

9.2 Limitations and Future Improvements

Current Limitations:

1. The knowledge base covers only three equipment categories and a limited number of diagnostic and inference rules.
2. The system depends on operator-entered qualitative symptoms and additional evidence.
3. Maintenance cost values are hypothetical estimates and are not based on verified supplier quotations.
4. Stakeholder weights and impact scores are illustrative and require real stakeholder validation.

5. Certainty factors are manually assigned and have not been validated using historical equipment data.
6. The backward-chaining mechanism performs limited hypothesis checking rather than general-purpose recursive reasoning.
7. The system does not automatically stop, control, or modify industrial equipment.
8. Diagnostic accuracy depends on the completeness and reliability of the knowledge base and user-entered evidence.

Future Improvements:

- Expand the knowledge base with additional equipment types and verified diagnostic rules.
- Validate maintenance cost estimates using actual maintenance records and supplier quotations.
- Collect stakeholder feedback to establish realistic weights and impact scores.
- Validate certainty factors using historical diagnostic data and expert assessments.
- Develop more comprehensive recursive backward chaining.
- Integrate industrial sensors and IoT monitoring for real-time evidence.
- Explore ontology-based knowledge representation and more advanced uncertainty-handling methods.

9.3 Conclusion

The project developed an Intelligent Knowledge-Based System prototype for Industrial Equipment Fault Diagnosis and Maintenance Decision Support.

The system represents industrial diagnostic knowledge through a structured rule-based knowledge base and applies forward chaining with additional evidence-based inference. It also provides limited backward-chaining hypothesis verification, certainty-factor support values, hypothetical maintenance cost estimates, stakeholder-based weighted priority ranking, safety alerts, and explanation facilities.

The implementation demonstrates how knowledge-based reasoning can support preliminary industrial fault diagnosis and maintenance decision-making

10. References

1. **Atlas Copco:** Air Compressor Troubleshooting Guide
2. **Haas Automation:** Air Compressor Operator's Service Manual
3. **LG&CD Punjab:** Water Supply System O&M Manual
4. **EASA:** Causes of Motor Overheating
5. **EASA:** Motor Vibration: Electrical or Mechanical

6. **EASA:** Determining Noise Sources in Electric Motors
7. **Sulzer Ltd.:** Vertical Pump Troubleshooting Guide
8. **Kaeser Compressors:** Troubleshooting Air Compressors
9. **Seoul Accord / NCEAC (2021):** Complex Computing Problem Attributes (CPA-2, CPA-3, CPA-9)